Simulator-Based Offline Reinforcement Learning for Sequential Movie Recommendation

Effects of Fatigue-Aware Reward Design on PPO and DQN

CS5180 Reinforcement Learning - Final Project Report

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Abstract

We study simulator-based offline reinforcement learning for sequential movie recommendation on MovieLens 1M. Our central question is whether reinforcement learning provides value beyond a myopic greedy recommender when the environment is learned from historical ratings rather than from live user sessions. The project proceeds in two stages. First, we analyze the real data and the learned simulator and show a mismatch: users diversify naturally, but the genre-level reward landscape is still highly exploitable. Second, we compare Random, Greedy-CTR, Deep Q-Network (DQN), and Proximal Policy Optimization (PPO) both without and with explicit fatigue-aware reward shaping. Without any artificial penalty, the policy ordering PPO > DQN > Greedy > Random is preserved across two independently trained GRU-based simulators, suggesting that reinforcement learning can exploit sequence-conditional dynamics even when a clear fatigue signal is absent. However, the no-penalty policies also collapse toward over-rewarded genres, indicating simulator bias. We therefore introduce window-based and concentration-based penalties that encode diversification directly into the reward. Under this fatigue-aware setting, PPO achieves the strongest gains over Greedy-CTR, learns a less concentrated policy than DQN, and benefits substantially from the genre-count state features. The main lesson is that in offline recommendation, reward design can matter as much as algorithm choice.

1. Introduction

Recommendation is inherently sequential. A choice that looks optimal at time step t can change what becomes attractive at time step t+1, t+2, and beyond. This creates a natural setting for reinforcement learning, which optimizes cumulative reward over an episode rather than one-step click probability alone. In practice, however, training directly on real users is costly and risky, so industry systems often rely on learned simulators for offline policy training before online deployment.

Our project uses MovieLens 1M as a controlled testbed for this workflow. The dataset is not a true session-based recommendation log, so we do not assume that it contains strong native fatigue behavior. Instead, we ask three narrower questions: (i) what structure is visible in the data itself, (ii) what kind of sequential advantage can reinforcement learning still obtain in a learned simulator, and (iii) how does explicit fatigue-aware reward shaping change policy behavior?

Our main contributions are threefold:

- We build an end-to-end offline reinforcement learning pipeline with genre-level actions, GRU-based user simulators, and policy evaluation against Random and Greedy-CTR baselines.
- We show that the no-penalty ranking PPO > DQN > Greedy > Random is robust across two simulators, but that these policies can exploit simulator bias and collapse toward a few over-rewarded genres.

• We demonstrate that explicit fatigue-aware reward shaping changes both performance and behavior: PPO improves the most, DQN remains more concentrated, and genre-count history becomes important once repetition is penalized.

2. Data, Simulators, and Markov Decision Process Setup

MovieLens 1M contains roughly 1 million ratings from 6,040 users over 3,706 movies spanning 18 genres. We binarize ratings by treating ratings of 4 or higher as positive feedback. The recommendation problem is formulated at the genre level rather than at the raw item level: at each step the agent chooses one of 18 genres, and the environment then selects the best currently available movie within that genre according to the simulator. This reduces the action space and makes policy behavior interpretable.

The state is 68-dimensional: a 50-dimensional user embedding plus an 18-dimensional vector of recent genre counts. Episodes last for 20 recommendation steps. The recent-genre window tracks the last 10 selections and is later reused for fatigue-aware reward shaping. We train and evaluate two GRU-based simulators. The original GRU4Rec model achieves area under the receiver operating characteristic curve (AUC) 0.703. The enhanced GRU4RecSVD model transfers a stronger sequential model from the companion CS6140 project and reaches AUC 0.781. For reference, a static XGBoost simulator attains AUC about 0.800, but it does not model sequence dynamics and is therefore not used as the main reinforcement learning environment.

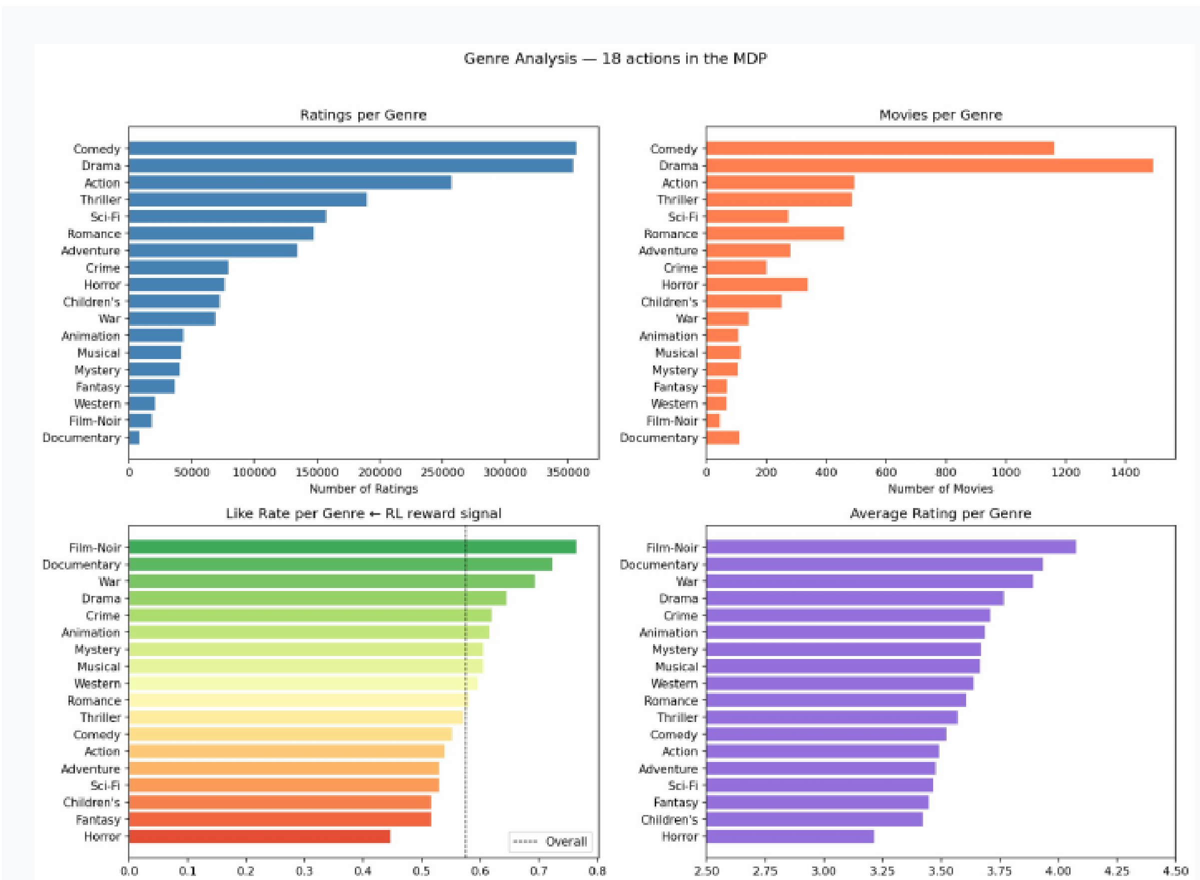

Fig. 1. Data insight plots used in the report: real users diversify across genres, yet genre-level like rates remain highly uneven, creating an exploitable reward landscape.

Figure 1 motivates the rest of the paper. On one hand, users are not naturally single-genre consumers: the reported median dominant-genre share is only about 24 percent, and user-level genre entropy is high. On the other hand, like rates vary sharply across genres, with Film-Noir among the most rewarded despite its small catalog size. This mismatch means that a policy can achieve high simulator reward by repeatedly targeting a few profitable genres even if that behavior is not representative of real viewing diversity.

Component	Specification
State	50-dimensional user embedding + 18-dimensional recent genre counts
Action	18 genre choices; environment picks the best available movie in that genre
Reward (no penalty)	Bernoulli(P(like)) from the simulator
Reward (fatigue-aware)	Bernoulli(clip(P(like) - penalty, 0.1, 0.9))
Episode length	20 recommendation steps
History window	10 recent genre selections

Table 1. Markov decision process design used for policy training and evaluation.

3. Real-Data Behavior Without Explicit Fatigue

Before adding any artificial penalty, we first ask whether the learned simulator already encodes a recoverable genre-fatigue signal. Under our genre-level probe, it does not. The mean predicted like probability remains nearly flat as same-genre repetition increases from k = 0 to 4, with only a very small change in magnitude. The key implication is that MovieLens does not naturally provide a strong repetition cost at the level we model here; any sequential advantage in the no-penalty environment must therefore come from other sequence-conditional structure in the simulator rather than from obvious fatigue.

Policy	Original GRU4Rec (AUC 0.703)	Enhanced GRU4RecSVD (AUC 0.781)	Rank preserved?
Random	10.05 ± 2.20	9.49 ± 1.78	Yes
Greedy-CTR	11.33 ± 2.43	10.88 ± 2.62	Yes
DQN	12.27 ± 3.12	12.25 ± 2.20	Yes
PPO	13.38 ± 2.79	12.63 ± 1.93	Yes

Table 2. No-penalty policy performance across the two GRU-based simulators. The same ordering is preserved in both environments.

Table 2 shows the most robust no-penalty result of the project: PPO remains the strongest policy, DQN is second, Greedy-CTR is third, and Random is last in both simulators. The PPO-DQN gap narrows in the stronger simulator, but the ordering does not change. This consistency suggests that the no-penalty advantage is not tied to one specific set of learned weights.

Why can reinforcement learning beat Greedy-CTR even when no clear fatigue signal is visible? The answer is that the simulator is still sequence-conditional. Greedy-CTR optimizes only the current step by choosing the genre whose best movie has the highest immediate predicted reward. PPO and DQN instead optimize the full 20-step episode and can learn genre orderings that alter future simulator states. At the same time, we should be cautious: because MovieLens is not a session dataset and because the simulator is imperfect, some of this advantage may reflect model-induced dynamics rather than genuine user behavior.

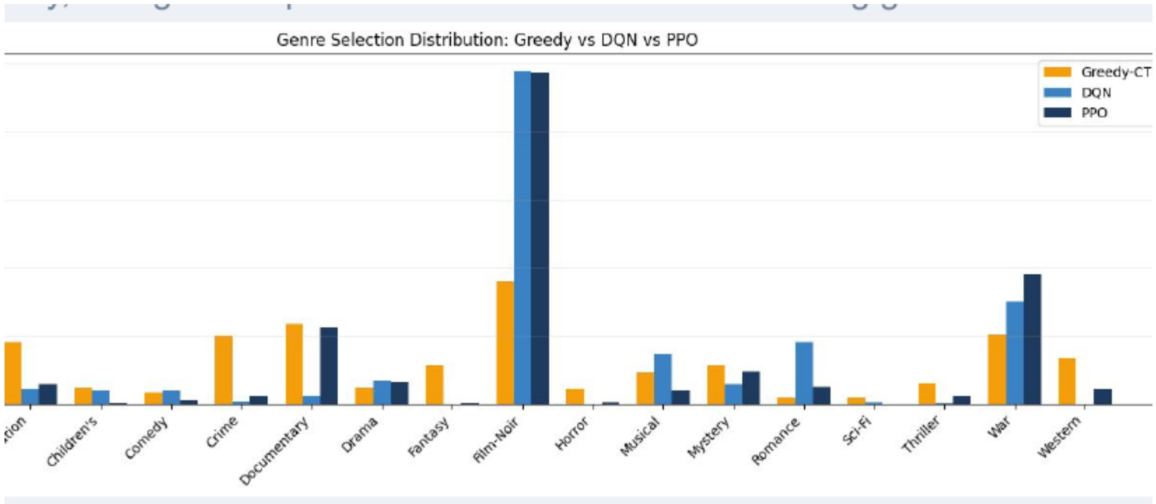

Fig. 2. No-penalty genre-selection distribution. Without a repetition cost, reinforcement learning policies collapse toward over-rewarded genres, especially Film-Noir, more strongly than the Greedy baseline.

Figure 2 reveals the main weakness of the no-penalty setup. The higher cumulative reward achieved by PPO and DQN does not imply realistic diversification. Instead, both policies heavily exploit simulator bias, allocating a disproportionate share of recommendations to a small set of genres. Greedy-CTR is more myopic and therefore less able to lock into the same kind of long-horizon exploitation. This observation motivates explicit reward shaping: if the data do not provide a strong fatigue signal, the environment may need one to align optimization with the desired behavior.

4. Fatigue-Aware Reward Design

To encode diversification directly, we introduce two penalties. The first is a short-horizon window penalty that increases with same-genre repetition within the last 10 steps: min(0.05 × same_count, 0.25). The second is a session-level concentration penalty that activates when a genre exceeds 25 percent of the picks made so far: 0.3 × max(genre_frac - 0.25, 0). These terms are subtracted from the simulator score before Bernoulli sampling. The goal is not to claim that MovieLens users truly behave according to these formulas, but to test whether reinforcement learning can respond to a reward function that explicitly values diversity.

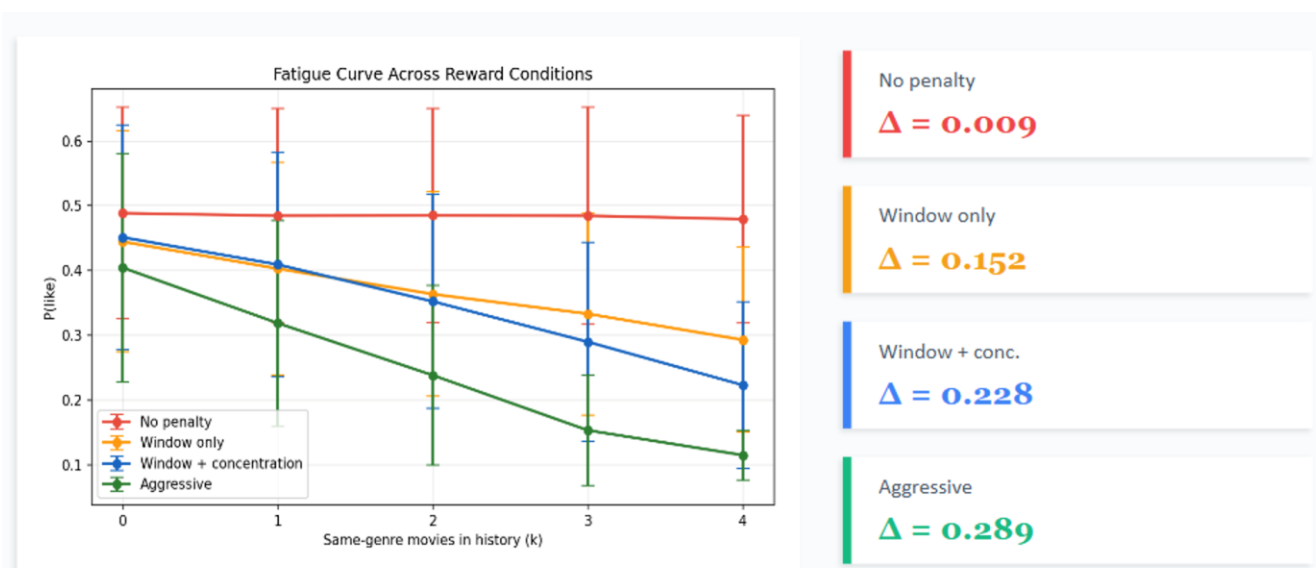

Fig. 3. Fatigue curves across reward conditions. Adding window and concentration penalties makes the effective reward decline more clearly as same-genre repetition increases.

Figure 3 confirms that the reward shaping behaves as intended. In the no-penalty condition, the curve is nearly flat. Under window-only and then window-plus-concentration penalties, the effective reward decreases more steeply as repeated exposure accumulates. This gives the agents a sequential pattern that Greedy-CTR cannot fully optimize because the immediate best action may create a worse future state.

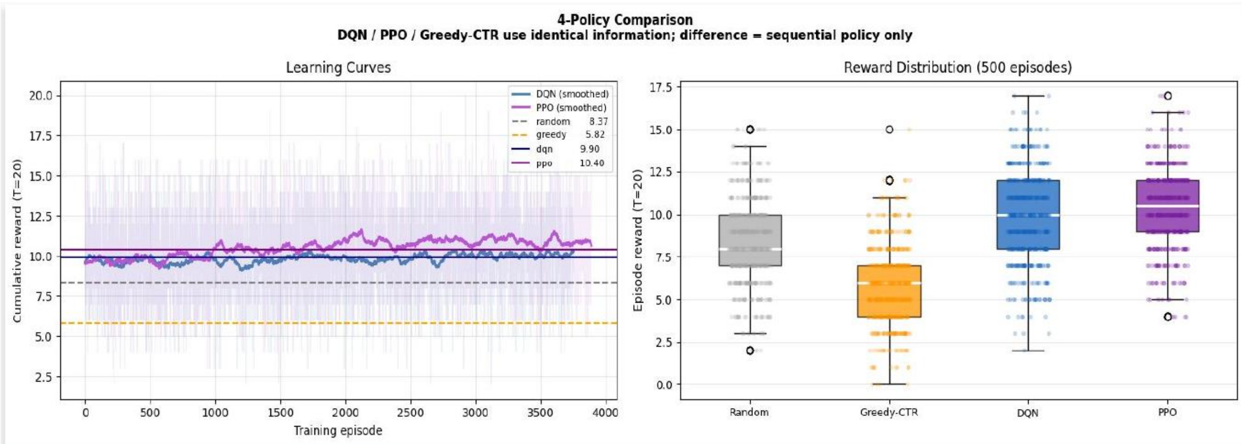

Fig. 4. Main fatigue-aware comparison under the window-plus-concentration reward. The learning curves and reward distributions show PPO clearly above DQN and Greedy-CTR.

Under the window-plus-concentration condition in Figure 4, PPO outperforms Greedy-CTR by +4.31 and DQN outperforms Greedy-CTR by +1.55, with both differences reported as statistically significant in the presentation results. This is the cleanest evidence in the project that sequential planning matters once the objective explicitly penalizes repetition. PPO also shows smoother improvement over training and a stronger reward distribution than DQN, which is consistent with PPO being better suited to this short-horizon, policy-sensitive environment.

5. Policy Behavior and Ablation Analysis

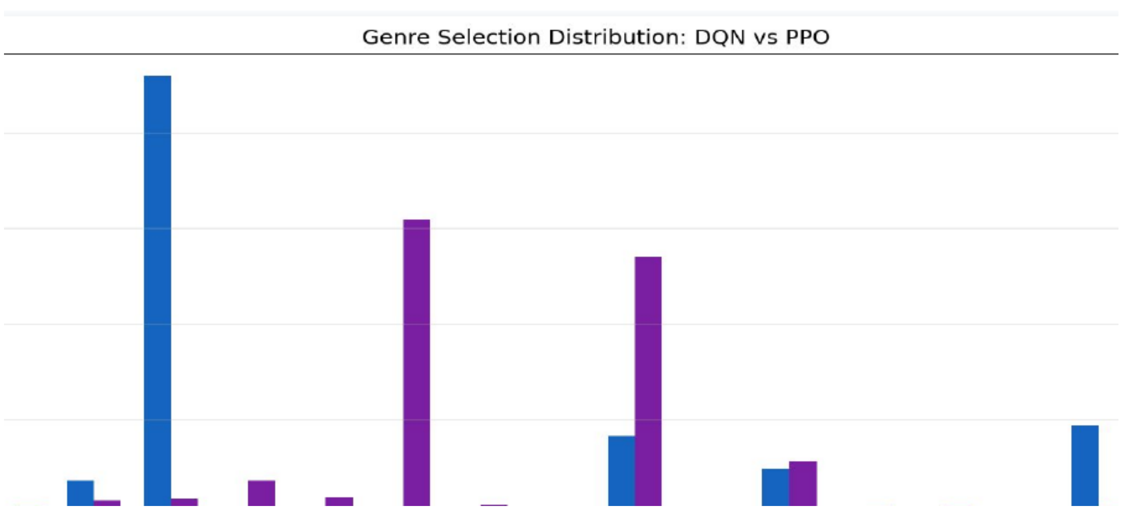

Fig. 5. Genre selection under fatigue-aware reward. DQN remains more concentrated, whereas PPO spreads mass across several genres and avoids a single-genre collapse.

Figure 5 helps explain why PPO wins. DQN still concentrates strongly on one genre, indicating that its deterministic action selection is more vulnerable to local exploitation even after penalties are introduced. PPO, by contrast, distributes its policy mass across multiple genres such as Documentary, Film-Noir, and War. In this project, the stochastic actor appears to provide a practical advantage: it explores and maintains a more balanced sequence strategy instead of converging to one dominant repeated action.

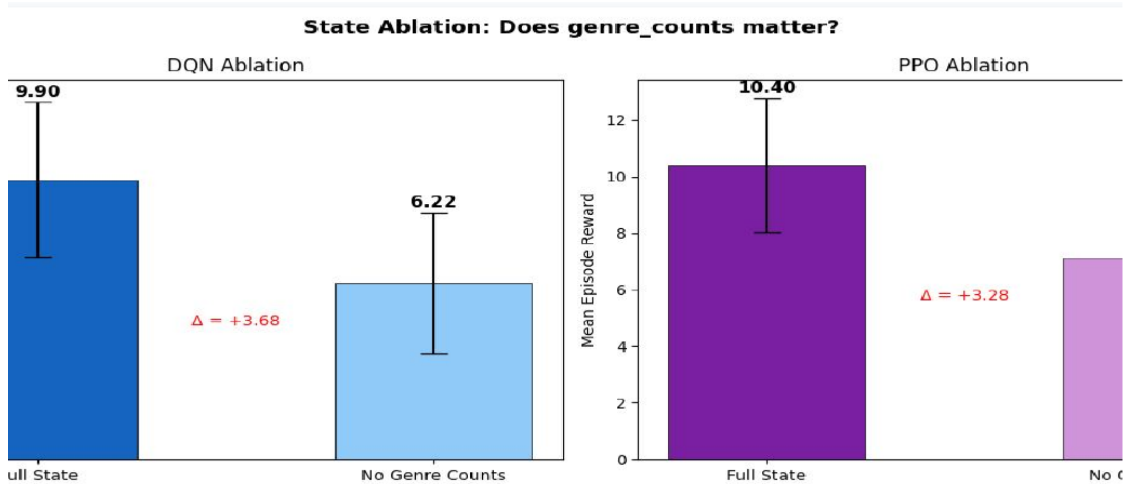

Fig. 6. State ablation under fatigue-aware reward. Removing the recent genre-count features sharply reduces performance for both DQN and PPO.

The state ablation in Figure 6 is also important. Once reward shaping depends on repetition and concentration, recent genre counts become an essential part of the state. Removing these 18 features reduces DQN from 7.22 to 5.17 and PPO from 9.98 to 5.55 in the reported experiment. This result contrasts with the no-penalty setting and shows that state design must match reward design: if the objective depends on history, the policy needs direct access to that history.

Appendix Figure A1 further compares all four reward conditions. The gap between PPO and Greedy-CTR grows as the penalty becomes stronger, while Greedy-CTR collapses rapidly when repeated exploitation is punished. This trend reinforces the broader conclusion that the environment definition - especially the reward - has first-order influence on what the learned policy becomes.

6. Discussion

The project yields two complementary messages. First, without an explicit fatigue term, reinforcement learning can still outperform a myopic greedy policy in a sequence model. That is an interesting empirical result, but it should be interpreted carefully because the advantage may come from simulator-specific sequence dynamics rather than from authentic user fatigue. Second, once we define a reward that explicitly discourages repetition, PPO and DQN both adapt, and PPO adapts best. In that setting, the reinforcement learning advantage is much easier to interpret because the desired long-term structure is built directly into the objective.

This distinction matters for the rubric as well. The strongest methodological contribution of the report is not the claim that MovieLens contains latent fatigue, but the more defensible claim that offline recommendation results depend jointly on simulator quality, reward design, and policy class. The no-penalty experiments test robustness across simulators; the fatigue-aware experiments test whether the pipeline can learn diversification when the task truly rewards it.

7. Limitations and Future Work

Limitations. MovieLens 1M is a static ratings dataset rather than a session log with watch time, skips, or dwell signals. Our genre-level fatigue probe therefore tests only one notion of repetition, and the simulator cannot be expected to recover rich session behavior from limited supervision.

Simulator bias. The no-penalty genre collapse demonstrates that offline reinforcement learning can exploit imperfections in the learned environment. A higher return in simulation is not sufficient evidence of better user experience.

Cold-start and representation limits. The current pipeline is still largely movie-identifier based. It does not solve true cold-start recommendation for new items or new users.

Future work. The most important next step is to move to genuinely session-based datasets such as KuaiRec or Tenrec, where fatigue-like effects are better aligned with the data-generating process. A second direction is to replace pure item identifiers with hybrid item features so that cold-start and generalization can be studied more realistically. Multi-seed evaluation and, eventually, online or human-in-the-loop validation would also strengthen the conclusions.

8. Conclusion

This report shows that simulator-based offline reinforcement learning can produce meaningful differences in sequential recommendation behavior even on a simplified movie domain. Across two simulators and two reward regimes, PPO is consistently the strongest method among the policies we test. Yet the project also shows why raw offline gains should be interpreted with care: without an explicit fatigue cost, the same algorithms can exploit simulator bias and learn unrealistic genre concentration. Once fatigue-aware reward shaping is introduced, the reinforcement learning advantage becomes both larger and more behaviorally aligned. The overall conclusion is therefore not simply that PPO beats DQN, but that reward design, simulator design, and state design must be treated as a coupled system.

References

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Appendix

A.1 Plot placement map

Figure	Plot to place	Purpose in report
Fig. 1	Data insight plots from the presentation	Motivate the mismatch between real diversification and exploitable rewards
Fig. 2	No-penalty genre distribution (Greedy vs DQN vs PPO)	Show simulator-bias exploitation without fatigue penalty
Fig. 3	Fatigue curves across reward conditions	Verify that the shaping terms create a usable sequential signal
Fig. 4	Window-plus-concentration learning curves and reward distributions	Main performance comparison under fatigue-aware reward
Fig. 5	DQN vs PPO genre-selection distribution	Explain why PPO is less concentrated than DQN
Fig. 6	State ablation: full state vs no genre counts	Show that history features matter when reward depends on history
Fig. A1	Reward shaping across four conditions	Supplementary trend showing PPO advantage grows with penalty strength

Table A1. Figure numbering and the plot images that should appear in the final white paper.

A.2 Additional reward-shaping comparison

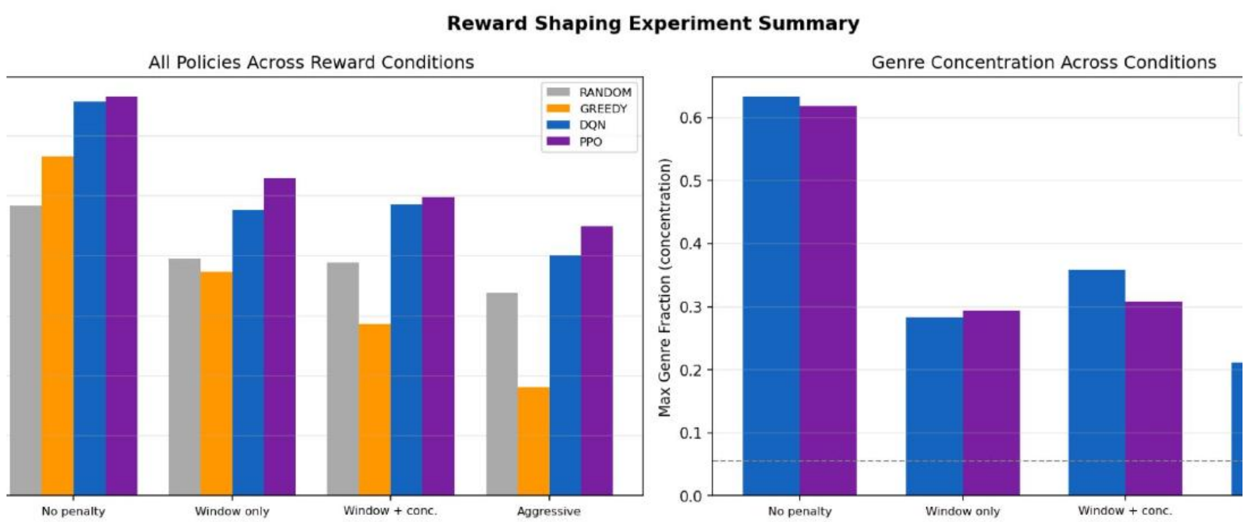

Fig. A1. Supplementary comparison across four reward conditions. PPO becomes relatively stronger as the repetition penalty becomes harder, while Greedy-CTR deteriorates quickly.